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Measurable Functions

1.1 Elementary Properties of Measurable Functions

Definition 1.1.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 1.1.2 ▶ Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable with respect to $\mathcal{M}$ and $\mathcal{N}$** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term “measurable mapping.”
- If Y is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the σ -algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When Y is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then f is called a **Borel measurable function**.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of f required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case f is called a **Lebesgue measurable function**. The definitions imply that E is a measurable set if and only if χ_E is a measurable function.

Theorem 1.1.3

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the σ -algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then Σ is a σ -algebra.

Proof. Easy to verify. □

Corollary 1.1.4

A continuous mapping is a Borel measurable function.

Proof. If f is continuous, Σ contains all open sets. Since Σ is a σ -algebra, Σ contains all Borel sets. □

Definition 1.1.5

If $f : X \rightarrow \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the **superlevel sets of f** .

Theorem 1.1.6

Let $f : X \rightarrow \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

- (i) f is measurable.
- (ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 1.1.7

A function $f : X \rightarrow \overline{\mathbb{R}}$ is measurable if and only if

- (i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
- (ii) $f^{-1}(a, b) \in \mathcal{M}$ for all open intervals $(a, b) \subset \mathbb{R}$.

Lemma 1.1.8

If f and g are measurable functions, then the following sets are measurable:

- (i) $X \cap \{x : f(x) > g(x)\}$,
- (ii) $X \cap \{x : f(x) \geq g(x)\}$,
- (iii) $X \cap \{x : f(x) = g(x)\}$.

Proof. If $f(x) > g(x)$, then there is a rational number r such that $f(x) > r > g(x)$. Therefore, it follows that

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g < r\}),$$

which is a measurable set, and easily (i) holds. (ii) is just the complements of (i) with f and g interchanged. (iii) is just the intersection two of two measurable sets of type (ii), and so it too is measurable. $\square$

Definition 1.1.9

If f and g are real-valued measurable functions, then $f + g$ is undefined at points where it would be of the form $\infty - \infty$. This difficulty is overcome if we define

$$(f + g)(x) := \begin{cases} f(x) + g(x), & x \in X - B, \\ \alpha, & x \in B, \end{cases}$$

where $\alpha \in \overline{\mathbb{R}}$ is chosen arbitrarily and where

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

Theorem 1.1.10

If $f, g : X \rightarrow \overline{\mathbb{R}}$ are measurable functions, then $f + g$ and fg are measurable.

Proof. Define $F : X \rightarrow \overline{\mathbb{R}} \times \overline{\mathbb{R}}$ by

$$F(x) = (f(x), g(x))$$

and $G : \overline{\mathbb{R}} \times \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$ by

$$G(x, y) = \begin{cases} \alpha, & \{x, y\} = \{\infty, -\infty\} \\ x + y, & \text{otherwise} \end{cases}$$

Where α is the value we determined where $f + g$ are taken on B ,

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

. Then $G \circ F(x) = f(x) + g(x)$. It suffices to show that $G \circ F$ is measurable. Referring to Theorem 1.1.7, we need first show that $(f + g)^{-1}(\infty)$ and $(f + g)^{-1}(-\infty)$ is measurable, which is a very easy but a little tedious process, thus we omit it here. We only need to show that $(G \circ F)^{-1}(J) \in \mathcal{M}$ whenever $J \subseteq \mathbb{R}$ is an open interval. Now $U := G^{-1}(J)$ is an open set in $\mathbb{R}^2$, since G is continuous on $\mathbb{R}^2$ and any point in $G^{-1}(J)$ cannot lie in $(\overline{\mathbb{R}})^2 \setminus \mathbb{R}^2$. Furthermore, U is the union of a countable family $\mathcal{F}$, of 2-dimensional intervals I of the form $I = I_1 \times I_2$, where I_1, I_2 are open intervals in $\mathbb{R}$. Since

$$F^{-1}(I) = f^{-1}(I_1) \cap g^{-1}(I_2)$$

We have

$$F^{-1}(U) = F^{-1}\left(\bigcup_{I \in \mathcal{F}} I\right) = \bigcup_{I \in \mathcal{F}} F^{-1}(I)$$

is measurable. Thus $G \circ F$ is measurable, since

$$(G \circ F)^{-1}(J) = F^{-1}(U)$$

The product is measurable by essentially the same proof. □